

DeepTutor



“ ” “ ”

aicourse.top / Agentic Engineering

2026-08-23 · DeepTutor v1.5.16 · commit 8515dfd...

NOT_ SCESS BLE ·

DeepTutor

Tutor

DeepTutor
AI Tutor RAG Memory
P0/P1/P2



1. “AI Tutor” Agentic
Engineering RAG Prompt Software Engineering
[L5][L7][L8]
2. AI Tutor “ ”
“ ”
3. RAG RAG BM25
/ / GraphRAG PageIndex
DeepTutor [D4][D5][D6]
4. DeepTutor “ ”
LLM judge [P1][P2]
5. “ ” Tutor
“ ” [L2][L3]
- “ Tutor DeepTutor ” RAG

1.

README manifest DeepTutor v1.5.16 README
AGENTS.md RAG Memory Mastery Question Book Sandbox DeepTutor v3 TutorBench
arXiv work-in-progress technical report
[D2][P1]

•	README		
•			
•	aicourse.top	DeepTutor	
			NOT_ASSESSABLE “available”“ ”“
”	[L1][L7]	DeepTutor	LLM judge 45 10
		[P2]	

2.

2.1

	NOT_ASSESSABLE	“ ” “ ”
	TOP_LEVEL_COURSES available	release gate
/	Cursor soon Claude Grok manifest	“ ” “Tutor ” [L7][L10]
	S00–S10	

DeepTutor A AI Agentic Engineering Codex
GitHub Prompts Software Engineering RAG MCP Make Money with Codex Claude Income
available README Available course “ ” [L1][L7]

- Next.js [L1][L2]
- localStorage [L3]
- RAG fail-closed [L5]
- Agentic Engineering harness RAG
capstone [L1][L5]

2.2 I Tutor

DeepSeek
Tutor [L3][L4]
RAG Retrieval Lab “ RAG” “top-k RAG” “ ” [L5][L6]

2.3

S00–S10 V3.1 Python LLM RAG AI Tutor
[L8]
Tutor policy Cursor Lesson 13 “ — —
” Cursor soon Tutor [L7][L9]

2.4

		DeepTutor v1.5.16		
	manifest QA	Book proposal spine typed blocks drift	Book	AI
Tutor	Tutor Cursor policy	/ pipeline	investigate-solve-write	
RAG	RAG	provider LlamaIndex BM25 vector RRF	SKG graph dense	
	localStorage learner model	L1/L2/L3 Memory Mastery	DPM Trace Forest schema	observation
	release checks Tutor outcome		LLM judge 45	→ →
/		provider		kill switch

DeepTutor

A

3. DeepTutor

3.1 “ ”

DeepTutor Chat Quiz Research Visualize Solve Mastery Path Immersive
Reading “ loop” Question Research Visualize Math Animator Book
pipeline BookEngine BaseCapability
[D1][D3]

3.2 R G

RAG factory provider LlamaIndex PageIndex Cloud PageIndex OSS GraphRAG
LightRAG LightRAG Server IMA provider MarginNote Obsidian capability/
factory provider LlamaIndex BM25 reciprocal-rank fusion
chunk id provider [D4][D5][D6]
needs_reindex version-N LlamaIndex embedding signature
provider [D4][D6] embedding

3.3 Memory

DeepTutor Memory L1 JSONL L2 Markdown L3
profile recent scope preferences Markdown L2 entity/trace ID
L3 /L2 L3 claim L2 entry id README “ ”
[D1][D7]

DPM Trace Forest / / dense embedding memory agents
L1/L2/L3 schema /
Memory [P2][D7]

3.4

DeepTutor Mastery Path
BKT IRT [D12]
Explore Plan Quiz /schema repair best-effort
issues validator / pipeline Evidence
Correctness Pedagogy schema [D8][P2]

3.5 Living Book

Book Engine proposal spine typed blocks
partial/error schema source_anchors
RAG [D1][D9]
“ ” fail-closed source-anchor —

4. DeepTutor

DeepTutor	Static Knowledge Grounding SKG				Dynamic Personal Memory DPM			
				Trace Forest		memory agents		
						v1.5.16		[P1][P2]
TutorBench	30	5	90	270	10	1–5	2	DeepTutor O
Q	3.91	Naive Tutor	3.53	+10.76%	Self-Refine	3.57	OQ	DeepTutor baseline
			SKG	groundedness/	/		DPM	[P2]
								DeepTutor
			Tutor					
		/LLM judge			Book Partners			
		personalization	SKG	DPM				TutorBench
harness	transcripts	judge records				[P1][P2]		

5. “ I Tutor ”

RAG Agentic Engineering S04/S05/S06/S08 Cursor Lesson 13
Tutor API Tutor [L8][L9]

1. Tutor		Agentic Engineering /	Tutor
2.		manifest RAG sources/checks	course-learning-contract.json
3. RAG		RAG 12 Retrieval Lab	
4.		Prompt Agentic loop	Tutor policy
5. Memory		context/memory	/
6.		quiz/capstone eval	Evidence/Correctness/Pedagogy gates
7. Tutor		Agent eval RAG eval	/
8.		Security Responsible AI	

DeepTutor A
capstone 50–100 retrieval gold queries 20–30
tutor dialogue cases protected outcome items / /

6. I Tutor

6.1

- **Explain**
- **Coach** → → → →
- **Practice** “ ”
Evidence/Correctness/Pedagogy P1-2b P0-1

6.2

fail-closed

Cursor

[\[L5\]](#)[\[L9\]](#)

6.3

” “ `concept_id` ” “ ” “ ” / “

6.4

Tutor “ ”“ ”“ ”“ ” /
“ / ”

7. R G

7.1

→ chunk → BM25 → → context packing → answerability gate →
 Retriever PageIndex/GraphRAG UI
 DeepTutor BM25 vector RRF
 [D5][D6]

7.2

Tutor-enabled

- course_id locale release_version content_hash
- lesson/page/section anchors
- rubric
- /provider embedding signature

“ ”
 fail closed

7.3

lesson / chunk “ ”“ ”“ ”
 answerability gate

7.4 R G

Recall@K MRR/nDCG / —
 p50/p95 / baseline
 authoring corpus Tutor development/eval cases
 judge rubric Tutor Tutor TutorBench judge Tutor
 ground truth [P2]

8.

8.1

Living Book Co-Writer typed block “ — — — — —
 ” DeepTutor anchor fail open [D9]

8.2

retrieve	read_source	grade	run_controlled_check	save_observation	Explain	Coach
Practice	Authoring				run_controlled_check	

8.3

Tutor			DeepTutor	/		token
	“	/	/	”		

8.4

						Memory
	SYSTEM/OS			[D10]		

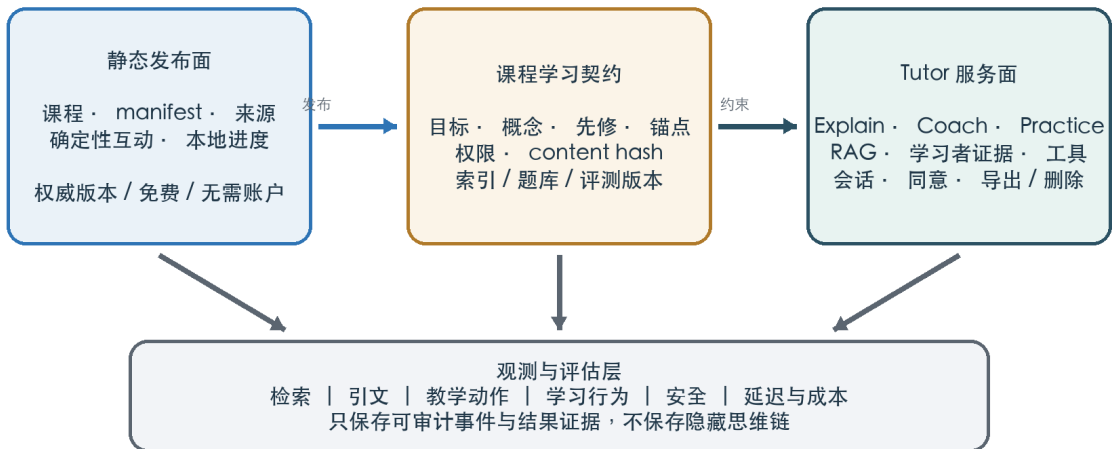
8.5

DeepTutor	Skills/MCP/CLI				“	”“	”“	”
	schema							

9.

推荐目标架构：双平面 + 课程学习契约

静态课程保持权威与低运维；Tutor 作为独立、可选、可撤销的服务运行。



1 Tutor

9.1

manifest	Next.js	Tutor	Tutor
[L2] [L3]			

9.2 Tutor

profile RAG / anonymous_session_id RAG

9.3

hash Tutor course_version contract_hash
fail closed Tutor feature flag kill switch

9.4

/ Tutor / authoring Tutor

10.

$$2 \quad 0.5 \quad / \quad /$$

P0-0		RAG Tutor cases	50-100 20-30 Tutor	/hash fail closed	RAG 3-4
P0-1	MVP	Explain Coach Practice	observation Memory	feature flag/kill switch /	/ 6-8
P1-2a				/	/ 4-6
P1-2b		Evidence/Correctness/Pedagogy gates	outcome set	/	10 6-
P1-3		proposal/spine/typed blocks S00-S10	diff	AI QA	Tutor 0 6-1
P2-4		PageIndex/		A/B /	

DeepTutor A

DeepTutor /

11.

11.1

“RAG ” Recall@K/MRR/nDCG / p95
/ baseline

11.2

rubric LLM judge
DeepTutor human alignment [P2]

11.3

“ ”“ ”“ Tutor”
/ “ ”

12.

			/
	DeepTutor	RAG SLO	/ Tutor
		/	
	QuestionPipeline schema r epair	Evidence/Correctness/Pedagogy gates	validator
	provider	A/B	
	Tutor /		
	retention/ /		[P2]
	Apache-2.0	LICENSE/NOTICE SBOM / / / /	[D11]

DeepTutor
A

OQ Naive Tutor 10.76% “ 10.76%” [P2]

13.

	RAG	50-100	20-30	
Tutor cases		Memory	/	/
		DeepTutor	/	

DeepTutor

- **[D1]** HKUDS/DeepTutor v1.5.16 [README](#) [Book KB Memory](#) [L203–213](#)
[Book](#) [typed blocks](#) [L568–578](#) [KB](#) [L583–595](#) [Memory L617–629](#)
- **[D2]** [v1.5.16](#)
- **[D3]** [Tools/Capabilities](#) [AGENTS.md L1–71](#)
- **[D4]** [provider](#) [KB](#) [embedding-signature](#) [RAG factory L1–75](#)
- **[D5]** [BM25](#) [RRF](#) [LlamaIndex retriever L119–151](#)
- **[D6]** [embedding signature](#) [needs_reindex](#) [LlamaIndex pipeline L126–226](#)
- **[D7]** [Memory L2](#) [L3](#) [trace.py L1–67](#) [document.py L1–33](#) [L64–102](#) [L3 p](#)
[rompt L1–37](#)
- **[D8]** [Explore/Plan/Quiz](#) [QuestionPipeline overview](#) [schema issue checks](#)
- **[D9]** [proposal spine typed blocks](#) [anchors](#) [best-effort RAG](#) [BlockType L59–83](#)
[SourceAnchor/Chapter L216–236](#) [Block anchors L411–425](#) [RAG](#) [L47–128](#)
- **[D10]** [SYSTEM/APPLICATION/OFF](#) [Sandbox spec L16–40](#)
- **[D11]** [Apache-2.0 LICENSE](#)
- **[D12]** [Mastery](#) [L61–223](#) [SQLite](#) [L1–45](#) [L13](#)
[–98](#) [L1–37](#)

DeepTutor

- **[P1]** [*DeepTutor: Towards Agentic Personalized Tutoring* v3 2026-07-09](#) [arXiv](#)
- **[P2]** [TutorBench](#) [arXiv v3 PDF](#)
- **[L1]** [README.md:1–7, 11–30, 72–120, 156–180](#) [README](#) / [RAG Agentic Engineering](#) /
- **[L2]** [next.config.ts:3–23](#) [output: "export"](#)
- **[L3]** [lib/progress.ts:1–52](#) [localStorage-only](#)
- **[L4]** [lib/deepseek.ts:1–11, 76–122](#) [components/lab/Lab.tsx:32–49, 85–142](#) [BYOK](#)

- [\[L5\]](#) lib/rag/manifest.ts:9–278 lib/rag/types.ts:48–126, 273–320 RAG 4 12 34 concepts 40 sources
- [\[L6\]](#) components/rag/RetrievalLab.tsx:8–79 RAG
- [\[L7\]](#) lib/courses.ts:290–450, 510–554 available Cursor soon
- [\[L8\]](#) course_review_2026-08-23/staging/AI _V3.1_ /manifest.json S00–S10
- [\[L9\]](#) messages/cursor/en.json:581–618 Cursor Lesson 13
- [\[L10\]](#) lib/claude/manifest.ts:158–164 lib/grok/course.manifest.json:1–30 app/[locale]/grok/page.tsx:1–18 lib/courses.ts:303–463 Claude rights-gated Grok manifest TOP_LEVEL_COURSES

B

- DeepTutor GitHub
- v3 PDF LLM judge
- 2026-08-23 NOT_ASSESSABLE available
- baseline /